

Experiments, Measurement, and Behavioral Identification in Labor

Behavioral Labor – Week 7

Teaching deck draft

Central question

- How do behavioral labor economists move from a behavioral object to an actual empirical method?
- Week 7 is the toolkit lecture: measurement, experiments, hazards, bunching, quasi-experiments, and structural estimation.
- The practical arc: object, labor margin, variation, estimand, method, interpretation.
- The recurring warning: a setting is not an estimand.

Bridge from Weeks 1–6

Earlier object	Week 7 empirical task
Preferences	recover dynamic, reference-dependent, or social preference parameters
Beliefs	elicit, validate, and link expectations to search and choices
Attention	separate salience, knowledge, and complexity from true incentives
Workplace behavior	estimate treatment effects and recover mechanisms
Identity and norms	distinguish measurement, sorting, treatment, and transmission

Week 8 will apply this toolkit to firm, market, and equilibrium response.

What counts as behavioral identification?

- Name the behavioral object: belief, preference, attention cost, reciprocity, norm sensitivity, or learning rule.
- Name the labor margin: search, exit, effort, take-up, applications, training, retention, or earnings.
- Name the variation: treatment, timing, threshold, schedule, exposure, survey panel, or model moments.
- Name the estimand: ITT, ATE, hazard effect, local elasticity, belief updating, or structural parameter.
- Name the strongest standard alternative.

Measurement: latent object and noisy proxy

$$z_{ij} = \theta_i + \eta_{ij}$$

Object	Interpretation
θ_i	latent behavioral object: belief, preference, attention, norm sensitivity
z_{ij}	survey response, elicitation, task behavior, admin proxy, or model moment
η_{ij}	measurement noise, demand effects, misunderstanding, timing mismatch

Behavioral labor often starts by asking whether the proxy really measures the object.

Behavioral object to data design

Object	Labor setting	Data object	Main challenge
Present bias	training, savings, effort	repeated choices, contract menus	preferences versus liquidity
Beliefs	job search	elicited expectations, linked outcomes	beliefs versus opportunities
Attention	schedules, take-up	admin records, notices, claims	salience versus constraints
Social preferences	workplace effort	real-effort output, gifts, wages	reciprocity versus selection
Norms	allocation, teams	coworkers, managers, surveys	treatment versus sorting

Field experiments in labor settings

$$Y_i = \alpha + \beta T_i + X_i' \gamma + \varepsilon_i$$

- Assignment identifies the treatment effect; OLS or ANCOVA is often the practical estimator.
- Baseline outcomes and strata improve precision.
- Randomization inference helps with small, clustered, or design-sensitive experiments.
- Heterogeneity matters when treatment works through prior beliefs, knowledge, constraints, or worker type.
- Mechanism interpretation still requires measurement.

Information interventions

Information object	Labor example	Interpretation issue
Belief level	job-finding probabilities	did expectations update?
Salience	EITC or benefit reminders	knowledge or attention?
Procedural knowledge	how to claim or apply	information or hassle reduction?
Confidence	perceived eligibility	optimism or actual opportunity?
Learning speed	schedule understanding	temporary or persistent response?

The regression may be simple; the behavioral claim is not.

Survey expectations and panels

- Repeated elicitation measures beliefs, bias, persistence, and updating.
- Panel fixed effects separate stable optimism or pessimism from within-person updating.
- Validation links beliefs to realized exits, wages, applications, or search effort.
- Forecast errors help distinguish beliefs from labor-market opportunities.
- Measurement error is central, not cosmetic.

$$h_{it} = \Pr(U_{i,t+1} = 0 \mid U_{it} = 1, X_{it}, B_{it})$$

- Search outcomes are often spells, exits, and censoring problems.
- Discrete-time logit or probit hazards fit period-by-period exit data.
- Cox-style hazards fit continuous duration settings.
- Competing risks matter when exits include jobs, training, inactivity, or benefit transitions.
- Behavioral question: beliefs, discouragement, learning, benefits, or opportunity?

Bunching, salience, and nonlinear schedules

$$\hat{e} = \frac{\Delta b/b}{\Delta(1-\tau)/(1-\tau)}$$

- Nonlinear tax, transfer, earnings, and bonus schedules create local variation.
- Bunching, kink, and notch methods estimate local response around schedule features.
- Low bunching may mean low elasticities, adjustment costs, inattention, or poor knowledge.
- Knowledge and salience data discipline behavioral interpretation.

Quasi-experiments in behavioral settings

Variation	Method family	Behavioral use
threshold or score	RD, kink, notch	eligibility, schedule knowledge, local response
staggered rollout	DiD, event study	information release, simplification, exposure
encouragement	IV	take-up when compliance is endogenous
policy timing	event study	learning, anticipation, dynamic adjustment
schedule reform	bunching, local elasticity	salience and incentive response

Structural estimation in behavioral labor

$$\hat{\psi} \in \arg \min_{\psi \in \Psi} \left[m^{data} - m^{model}(\psi) \right]' W \left[m^{data} - m^{model}(\psi) \right]$$

- ψ can include present bias, reciprocity, search costs, learning rules, or attention costs.
- MLE, SMM, dynamic discrete choice, and calibrated moments are common tools.
- Structural work is useful for parameter recovery and counterfactuals.
- The price is model dependence: state which moments identify which parameters.

Practical econometric methods cheat sheet

Empirical setting	Common method	Add-ons
Randomized labor intervention	OLS, ANCOVA	randomization inference, heterogeneity
Repeated beliefs panel	fixed effects, validation	forecast errors, measurement error
Job-search duration	logit/probit hazards, Cox	competing risks, dynamic controls
Nonlinear schedule	bunching, local elasticity	knowledge interactions
Policy or information shock	IV, RD, DiD, event study	spillovers, dynamics
Parameter recovery	MLE, SMM, dynamic model	fit, sensitivity

Heterogeneity is a method choice

- Behavioral effects often vary by baseline beliefs, knowledge, attention, liquidity, identity, or workplace context.
- Interactions are transparent when the moderator is pre-treatment.
- Random coefficients or hierarchical models can summarize richer heterogeneity.
- Machine-learning heterogeneity tools can be useful, but the behavioral object still needs interpretation.
- Never let heterogeneity replace a mechanism.

Common failure modes

Failure	Diagnostic question
Setting without estimand	what parameter is estimated?
Beliefs versus opportunities	are expectations validated against outcomes?
Treatment versus measurement	what information object actually moved?
Bunching overreach	preferences, knowledge, or adjustment costs?
Structural overreach	which moments identify which parameters?
Welfare over-claiming	behavior only, or welfare too?

- Reproduce: synthetic job-search information experiment anchored on Altmann–Falk–Jaeger–Zimmermann.
- Diagnose: behavioral object, margin, variation, method, assumptions, interpretation.
- Transfer: workplace gift exchange and social-preference recovery.
- Challenge: nonlinear schedule response or self-control at work.

- Week 7 supplies the empirical toolkit.
- Week 8 asks what survives once worker-level behavioral wedges enter firms, markets, search, and sorting.
- The method discipline separates worker response, firm strategy, and market outcomes.
- Week 9 then turns that equilibrium view into behavioral public policy design.

Takeaways

- Behavioral labor combines standard labor methods with behavioral measurement and modeling.
- Experiments estimate treatment effects; mechanisms require measurement.
- Beliefs panels call for validation and fixed effects.
- Search durations call for hazards; nonlinear schedules call for bunching and local elasticities.
- Structural tools are for parameter recovery and counterfactuals, not decorative language.